

EECS 4201 Stage 2 Synthesis and Timing Report

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Design: 5-stage in-order RV32I core, top module rv_core

Date generated: October 6, 2026

Summary

Item	Result
Synthesis status	Yosys completed and generated synth.v and synth.json
Target cell library	sky130_fd_sc_hd, 130 nm standard-cell library
Total chip area	479,444.8256 um ²
Sequential area	287,615.8464 um ² (59.99%)
Mapped cells	48,815
ABC timing estimate	clk_period_min = 5.401 ns, fmax = 185.17 MHz

Synthesis preparation

The updated synthesis script was placed in the synthesis directory and adapted to the project folder layout. The Stage 2 design files forwarding_unit.sv, hazard_detection_unit.sv, and control_pipeline.sv are included in the synthesis file list. For synthesis only, main_memory.sv uses a fixed MEM_BYTES value of 1024 and excludes simulation-only program loading and debug constructs with SYNTHESIS guards. Normal Verilator simulation behavior remains outside those guards.

Area and structure statistics

Metric	Value
Wires	54,945
Wire bits	88,105
Public wires	542
Public wire bits	12,279
Ports	3
Port bits	3
Cells	48,815
Chip area	479,444.8256 um ²
Sequential element area	287,615.8464 um ² (59.99%)

Largest cell types by total area

Cell type	Count	Area (um ²)	Share of chip area
edfxtp_1	9,216	277,000.0	57.78%
o21ai_0	7,827	39,200.0	8.18%
o211ai_1	3,584	26,900.0	5.61%
nor2_1	5,652	21,200.0	4.42%
nand2_1	5,029	18,900.0	3.94%
a22oi_1	2,020	15,200.0	3.17%
nand3_1	3,043	15,200.0	3.17%
inv_1	3,521	13,200.0	2.75%

Most frequent cell types

Cell type	Count	Area (um^2)
edfxtp_1	9,216	277,000.0
o21ai_0	7,827	39,200.0
nor2_1	5,652	21,200.0
nand2_1	5,029	18,900.0
o211ai_1	3,584	26,900.0
inv_1	3,521	13,200.0
nand3_1	3,043	15,200.0
clkbuf_1	2,552	9,580.0

Timing estimate

ABC reported a mapped critical path delay of 5,400.55 ps after buffering and sizing. This corresponds to `clk_period_min` = 5.401 ns and `fmax` = 185.17 MHz. The synthesis script also invokes the provided OpenSTA timing.tcl hook. On this laptop, the sta binary was not installed, so this report records the ABC timing estimate from the completed synthesis log. If OpenSTA is available on the grading machine, the same synthesis script will invoke timing.tcl.

Field	Value
ABC start point	pi99 (\mem_wb_rd [3])
ABC end point	po2130 (\$auto\$rtlil.cc:3497:MuxGate\$58443)
Estimated minimum clock period	5.401 ns
Estimated fmax	185.17 MHz

Interesting findings

- Sequential elements consume 59.99% of total mapped area, which is expected for a pipelined design with pipeline registers, register file storage, and memory/register structures.
- The largest area contributor is the enabled flip-flop cell `edfxtp_1`, used heavily by state-holding logic.
- Frequent combinational cells such as `o21ai_0`, `nor2_1`, `nand2_1`, `nand3_1`, and `o211ai_1` come from control decoding, muxing, forwarding comparisons, and datapath selection logic.
- The critical path reported by ABC ends at mux-generated logic, which is consistent with forwarding/control selection and writeback datapath muxing in a pipelined core.

Raw synthesis log excerpts

These excerpts are copied from `synthesis/yosys_stage2.log`, the system-generated Yosys output produced by running the synthesis script.

```
yosys> stat -liberty synthesis/sky130nm.lib

34. Printing statistics.

=== rv_core ===

54945      - wires
88105      - wire bits
  542      - public wires
12279      - public wire bits
   3       - ports
   3       - port bits
48815 4.79E+05 cells

Chip area for module '\rv_core': 479444.825600
of which used for sequential elements: 287615.846400 (59.99%)
ABC: WireLoad = "none"  Gates = 39072  Area = 191828.97  Delay = 5400.55 ps
ABC: Start-point = pi99 (\mem_wb_rd [3]).
ABC: End-point = po2130 ($auto$rtlil.cc:3497:MuxGate$58443).

Estimated from ABC:
clk_period_min = 5.401 ns
fmax = 185.17 MHz
```

Generated artifacts

- synthesis.log

- synth.v
- synth.json
- synthesis/yosys_stage2.log
- synthesis/stage2_synthesis_timing_report.pdf